CODEALPHA DATA SCIENCE INTERNSHIP — TASK 2

Unemployment Analysis in India

Exploratory Data Analysis and COVID-19 Shock Assessment

> **National unemployment surged from 9.5% to 17.8% during the COVID shock window, representing an 86.9% increase in reported joblessness across 28 regions.**

Source: India Unemployment Dataset (May 2019 – June 2020)

From raw upload to analysis-ready data

1 **Format discovery & ingestion**

Identified .xls file as UTF-8 CSV with BOM. Performed ingestion using Python's Pandas with explicit encoding handling.

2 **Cleaning & structural validation**

Removed 28 blank records; parsed dates using day-first convention; coerced numeric fields to float64; stripped whitespace from region/area labels.

3 **Analytical enrichment**

Categorized observations into pre-COVID and shock windows; derived monthly periods for time-series aggregation.

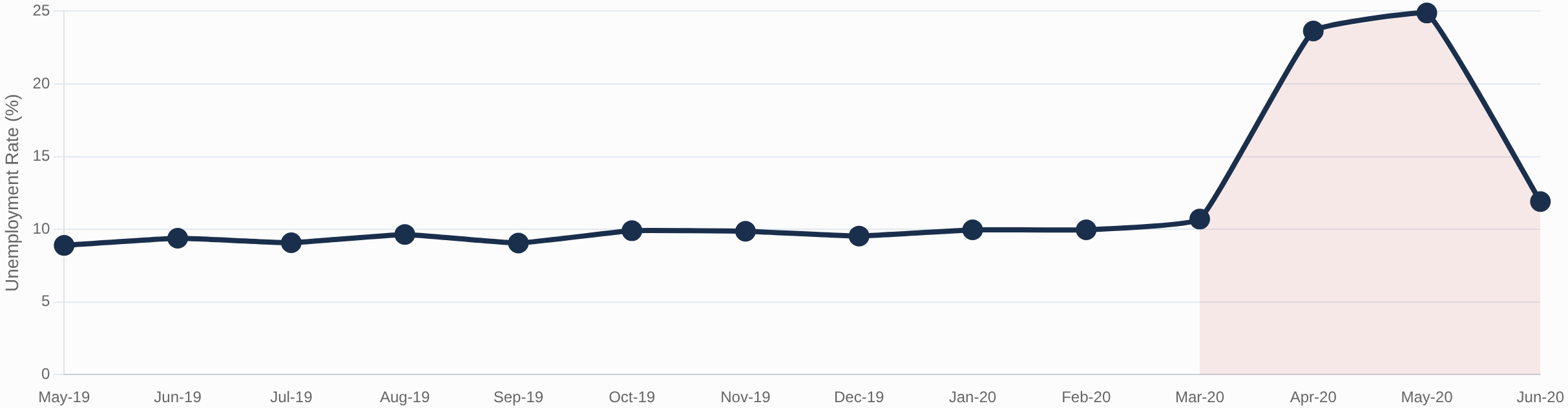
Period definitions

Period	Date Range	Purpose
Pre-COVID	May 2019 – Feb 2020	Baseline reference
COVID shock	Mar 2020 – Jun 2020	Assessment window
Full Sample	May 2019 – Jun 2020	Trend analysis

> The final analysis-ready dataset contains **740 high-fidelity observations** across **28 regions** and **2 area types** (Urban/Rural).

Source: India Unemployment Dataset (May 2019 – June 2020)

The COVID shock dominates the time series



24.88%

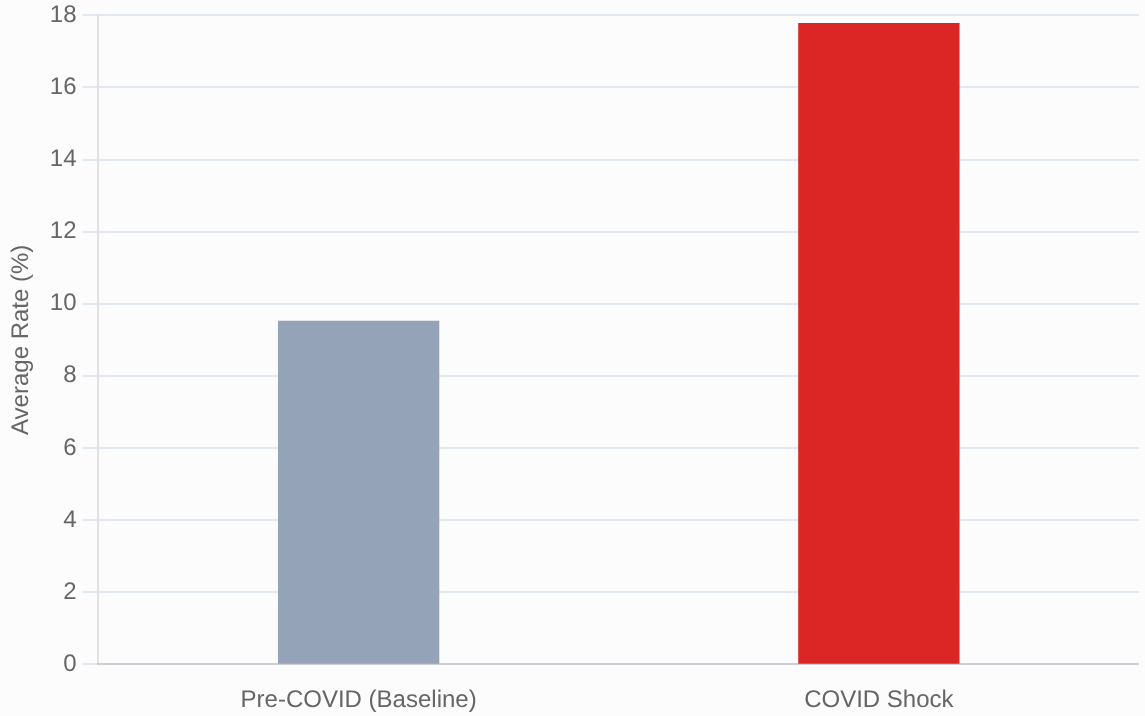
NATIONAL PEAK (MAY 2020)

The national average surged in April 2020 as lockdown measures took effect, peaking in May before a slight correction in June. The transition from a stable 9% baseline to 24% occurs in just 60 days.

Source: India Unemployment Dataset (May 2019 – June 2020)

Average unemployment rose 86.9%

Pre-COVID vs. Shock Period Average



Area-specific performance

+77.8%

URBAN INCREASE

Baseline: 10.8%
Shock: 19.3%

+99.9%

RURAL INCREASE

Baseline: 8.1%
Shock: 16.2%

Rural areas experienced a higher relative surge in unemployment (+99.9% vs +77.8% in urban zones), highlighting a widespread geographic shock that outpaced urban resilience.

Source: India Unemployment Dataset (May 2019 – June 2020)

Regional averages hide a wide spread

Region	Avg Rate (%)	Peak Rate (%)
Tripura	**28.35**	**43.64**
Haryana	**26.28**	**46.89**
Jharkhand	**20.58**	**61.48**
Bihar	**18.92**	**58.77**
Himachal Pradesh	**18.54**	**50.00**
Delhi	16.50	45.78
Jammu & Kashmir	16.19	24.06
Chandigarh	15.99	22.03
Rajasthan	14.06	24.15
Uttar Pradesh	12.55	26.43
Punjab	12.03	28.33
Puducherry	10.22	76.74

*Top 12 regions sorted by average rate.

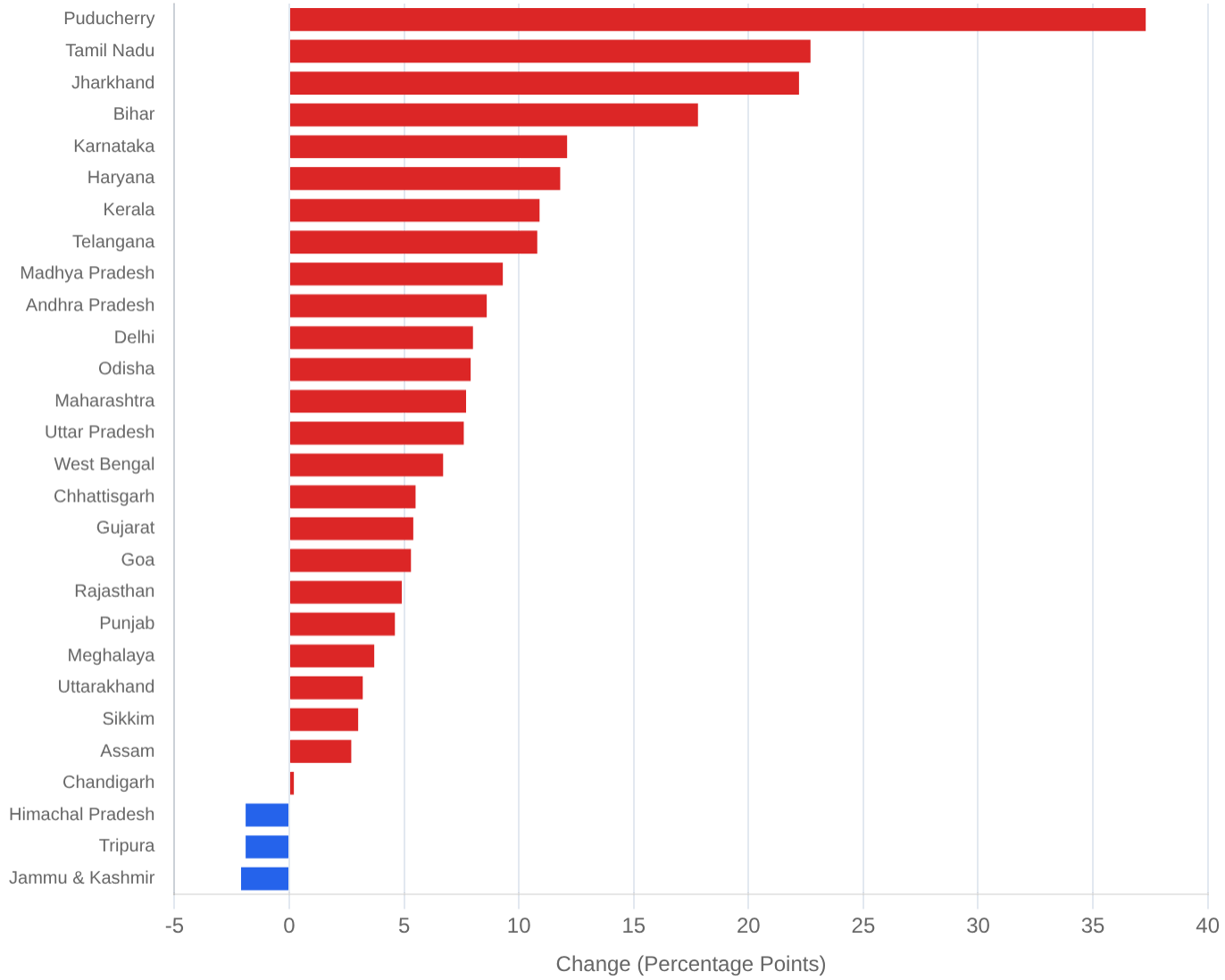
Key observation

National unemployment is a composition of vastly different regional outcomes. The top five regions averaged **18.5% to 28.4%**, while regions like Meghalaya (4.8%) and Odisha (5.6%) maintained significantly lower averages throughout the period.

Extreme Outlier:
Puducherry recorded a **76.7% peak** during the COVID shock, illustrating how specific territories experienced complete labour-market shutdowns.

Source: India Unemployment Dataset (May 2019 – June 2020)

COVID effects were uneven across regions



Geographic Variance

Extreme Shock: Puducherry and Tamil Nadu saw average rate increases of **20-35 percentage points**, reflecting direct exposure to lockdown intensity.

Resilience: Jammu & Kashmir and Tripura actually saw slight decreases in their average rates during this specific window, suggesting different local economic drivers or data reporting lags.

Broad Stress: The majority of regions (24 of 28) saw an absolute increase in average unemployment.

This uneven impact confirms that **tailored regional responses** are more efficient than uniform national policies.

Source: India Unemployment Dataset (May 2019 – June 2020)

What the evidence suggests for policy

1. Target by geographic stress

Resource allocation for unemployment insurance and public works should prioritize high-average and high-increase regions. A blanket national strategy misses the intensity of the shock in states like Jharkhand, Bihar, and Tamil Nadu.

2. Differentiate urban vs. rural

Rural areas saw a near 100% surge in unemployment, while urban areas faced higher absolute levels but slightly lower relative increases. Recovery strategies must address the distinct transmission mechanisms in both sectors.

3. Monitor participation flows

Policy should monitor labour force participation alongside unemployment. A falling unemployment rate can be a false signal of recovery if it is driven by workers discouraged into permanent exit from the labour market.

4. Invest in high-frequency data

The speed of the COVID shock (baseline to peak in 60 days) demonstrates the need for monthly regional monitoring. Annual or quarterly national statistics are too slow for emergency labor-market management.

Source: India Unemployment Dataset (May 2019 – June 2020)

Takeaways and next steps

Final assessment

Structural Shock: National average unemployment surged **+86.9%** during the initial COVID shock, peaking in May 2020.

Dual Sector Stress: Rural unemployment nearly doubled relative to its baseline, while urban areas faced high absolute levels (19.3%).

Regional spread: A wide variance from Tripura (28.4% avg) to Meghalaya (4.8% avg) confirms that national averages mask local stress.

Next steps

1. Extend dataset through 2021 to assess the full recovery trajectory.

2. Correlate findings with sector-specific GDP data by region.

3. Incorporate wage and inflation data for a real-income analysis.

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Source: India Unemployment Dataset (May 2019 – June 2020)